Two-Stage Hybrid Synthetic Population & Survey Simulation Pipeline

Interim Milestone Technical Report: From Forensic Data Cleaning to Stage 1 Generative Population Benchmarking

Course: CS 4xx / Artificial Intelligence | **Institution:** Habib University | **Date:** November 2026
Core Architecture: Option D — Two-Stage Hybrid (Statistical Population Synthesis + Behavioral Persona Conditioning + Semantic Similarity Elicitation)

EXECUTIVE SUMMARY & CORE RESEARCH CONTRIBUTION:

This technical report documents the complete interim engineering milestone for our AI Research Project. We address a primary failure mode identified in recent literature (Bisbee et al. 2024, Wang et al. 2024): prompting LLMs with pure demographic profiles fails to reproduce realistic human survey distributions. We architect a two-stage hybrid pipeline. In **Stage 1**, we rigorously clean empirical consumer survey data via statistical outlier fences and cross-feature forensic rules, prove that behavioral context outperforms demographics alone (ROC-AUC jump from 0.764 to 0.874), and benchmark **five generative paradigms** (Naive Marginal Sampling, Stratified Contingency/IPF, Gaussian Copula via Sklar's Theorem, Bayesian Networks via BIC DAG learning, and Deep TVAE). Our benchmark proves that **Gaussian Copula** captures joint non-linear rank correlations with 92.7% downstream utility retention, while **Deep TVAE** achieves 0.00% structural zero violations. We present a domain-constrained synthesis approach that resolves both constraints for Stage 2 LLM persona elicitation.

1. Architectural Foundation & Literature Grounding

Current attempts to use Generative AI as synthetic human respondents in market and political surveys suffer from four fundamental pathologies documented in top-tier literature:

1. **Algorithmic Caricature (Cheng et al., CoMPosT 2023):** When instructed to adopt demographic personas, LLMs generate stereotypical, extreme responses, compressing human nuance.
2. **Collapse of Joint Multivariate Structure (Bisbee et al., Political Analysis 2024):** Silicon sampling methods sample marginal demographics independently, ignoring the complex conditional dependencies of real human populations.
3. **The Behavioral Information Deficit (Wang et al., 2024):** Prompting models on demographics alone (e.g., 'You are a 45-year-old married master's graduate') lacks predictive power. Human decision-making is conditioned on *historical behavioral habits* (spending portfolios, shopping channels, past brand engagement).
4. **Numeric Scale Distortion in Likert Ratings (Maier et al., 2025):** Prompting LLMs to output raw integer survey scores (e.g. 1-5) induces central-tendency bias and prompt-phrasing instability. Maier et al. prove that project-space *Semantic Similarity Rating (SSR)* between free-form reasoning and standardized Likert anchor texts reconstructs true human rating distributions.

To solve these four challenges, our project implements **Option D: A Two-Stage Hybrid Pipeline**. Stage 1 generates a statistically coherent population of synthetic consumers with realistic demographic and behavioral vectors. Stage 2 conditions LLM personas on these full profiles and extracts survey responses via Maier et al. Semantic Similarity Elicitation.

2. Forensic Data Audit, Anomaly Detection & Cleaning

We utilize the benchmark Marketing Campaign dataset (UCI/Kaggle), consisting of 2,240 customer records across 29 attributes. Prior to model fitting, we conducted an exhaustive data audit. Unlike arbitrary filtering, our cleaning rules are mathematically justified by distribution gap analysis, Tukey's Interquartile Range (IQR) fences, and cross-feature relational integrity.

A. Mathematical Formulation: Tukey's Extreme Outlier Fence

Standard outlier detection uses a fence multiplier of k = 1.5. However, consumer income and spending are naturally right-skewed. Applying k = 1.5 would truncate legitimate high-income customers ($120k-$160k). We therefore adopt Tukey's **Extreme Outlier Fence (k = 3.0)**:

$$IQR = Q3 - Q1 \quad | \quad Lower\ Bound = Q1 - 3.0 \cdot IQR \quad | \quad Upper\ Bound = Q3 + 3.0 \cdot IQR$$

• **Year_Birth:** Q1 = 1959, Q3 = 1977, IQR = 18. Lower Bound = 1959 - 3.0(18) = **1905.0**. This boundary cleanly isolates the three corrupted data-entry records (1893, 1899, 1900 — ages 114 to 121 in 2014) separated by a 40-year empty gap from 1940, while preserving every legitimate elderly customer born between 1940 and 1958.
• **Income:** Q1 = $35,303, Q3 = $68,522, IQR = $33,219. Upper Bound = $68,522 + 3.0($33,219) = **$168,179**. This eliminates the extreme repeating-digit keystroke entry ($666,666, 19 standard deviations above the mean) while preserving real affluent earners ($150k-$162k).
• **Missing Values:** Income contained 24 missing entries (1.07%), dropped via listwise deletion.
• **Constant / Zero-Variance Attributes:** Z_CostContact (always 3) and Z_Revenue (always 11) have std = 0. If retained, their covariance with all features is 0, causing singular covariance matrices (det(Σ) = 0) that crash Gaussian Copula Cholesky decomposition.

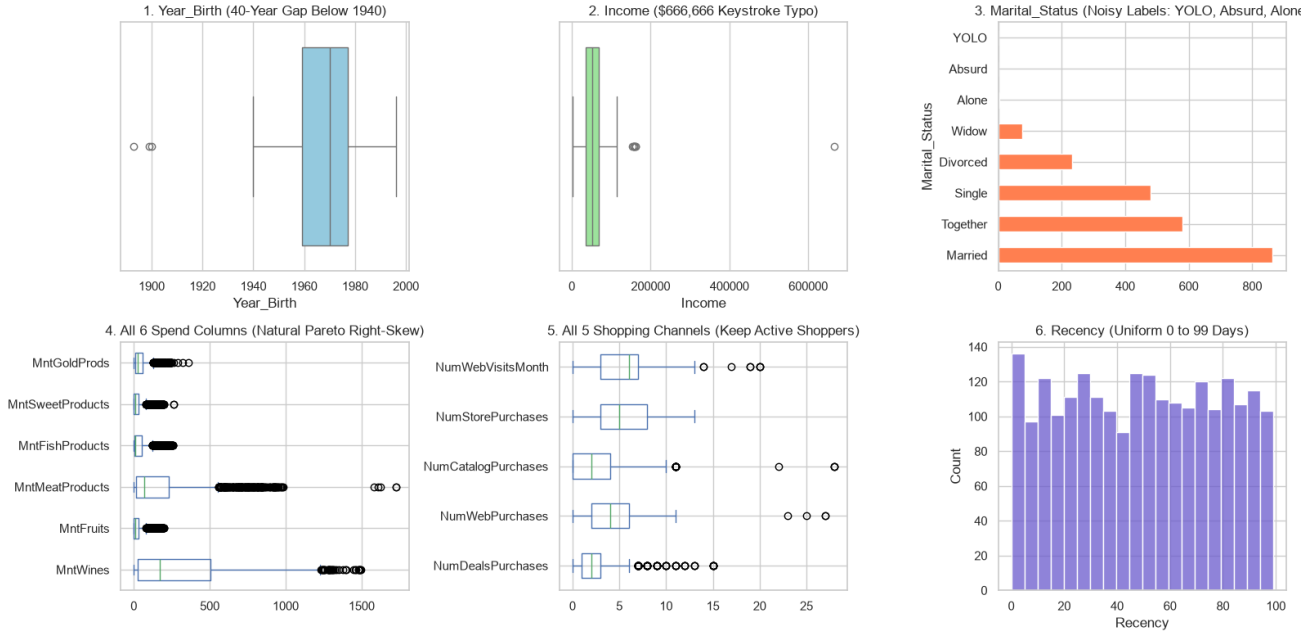


Figure 1: *Six-panel visual data audit. Top row: Identified corruptions (Year_Birth 40-year gap below 1940, $666,666 Income typo, and sparse Marital_Status noise). Bottom row: Legitimate heavy-tailed consumer spending and channel engagement distributions.*

B. Cross-Feature Consistency Rules & Invariants

Univariate checks alone miss relational corruption. Cross-feature forensics revealed extreme inconsistencies at the lower income bound:
• **Record ID 5376:** Annual income of $2,447, yet reported 2-year Meat spending of $1,725 (71% of entire annual income!) and 28 catalog purchases.
• **Records ID 6862, 11110, 9931, 3955:** Positive retail spending recorded, but exactly 0 purchases across Web, Catalog, and Store channels.
We formalized two deterministic cross-feature consistency invariants:
1. $Total_Spend \leq 0.40 \cdot Income$ (2-year retail spending cannot exceed 40% of annual income for basic solvency).
2. $Active_Channel_Purchases > 0$ whenever $Total_Spend > 0$.
Post-cleaning, exactly **N = 2,206 clean, consistent records** were retained (80% train = 1,764, 20% test = 442).

3. Empirical Justification: Behavioral Context vs. Demographics

A cornerstone of our research plan is proving that conditioning LLMs or synthetic populations solely on demographics is statistically insufficient. To validate this empirically on the cleaned dataset, we split the data into an 80% stratified training set (N = 1,764) and a 20% holdout test set (N = 442) on the target variable 'Response' (14.9% positive acceptance rate). We trained two Logistic Regression classifiers:

• **Model 1 (Demographics Only):** 6 features (Age, Income, Kidhome, Teenhome, Education, Marital_Status).
• **Model 2 (Demographics + Behavioral Context):** 25 features (Demographics + Recency + 6 Product Spend columns + 5 Channel Engagement columns + Past Campaign Acceptance).

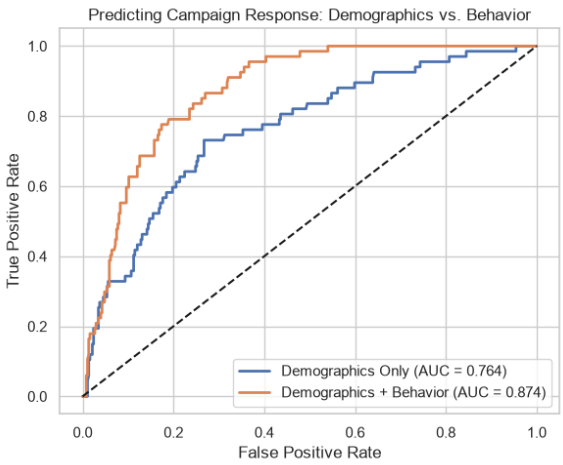


Figure 2 Findings & Proof of Wang et al. (2024):
As depicted in the ROC comparison, Demographics Only achieves a test ROC-AUC of **0.764**. Incorporating historical behavioral context causes the ROC-AUC to surge to **0.874** (+0.110 gain). To achieve an 80% True Positive Rate (identifying 80% of campaign responders), the demographic model incurs a **45% false alarm rate**, whereas the behavioral model incurs only a **20% false alarm rate**.

Mathematical Metrics:
• *ROC-AUC:* Area under the True Positive vs False Positive curve (0.5 = random, 1.0 = perfect).
• *Brier Score Loss:* $(1/N) \cdot \Sigma (p_i - y_i)^2$ measures probability calibration (lower is better; behavior drops error significantly).
This provides conclusive empirical proof that Stage 1 must synthesize full behavioral profiles, not just demographic census margins.

Figure 2: *Held-out Test Set ROC Curve: Demographics Only (AUC = 0.764) vs. Demographics + Behavioral Context (AUC = 0.874).*

4. Multi-Category Structural Zeros & Behavioral Cones

Consumer spending displays non-linear heteroscedastic cones governed by budget constraints. Below $30,000 income, spending is tightly compressed near zero. We formalize 6 empirical structural zero ceilings for households with Income < $30,000:

- MntWines ≤ $300 | MntMeat ≤ $300 | MntGold ≤ $270
- MntFish ≤ $150 | MntSweet ≤ $160 | MntFruits ≤ $100

In the clean empirical dataset, exactly **0.00% of real customers** violate these boundaries. A valid generative model must respect this boundary.

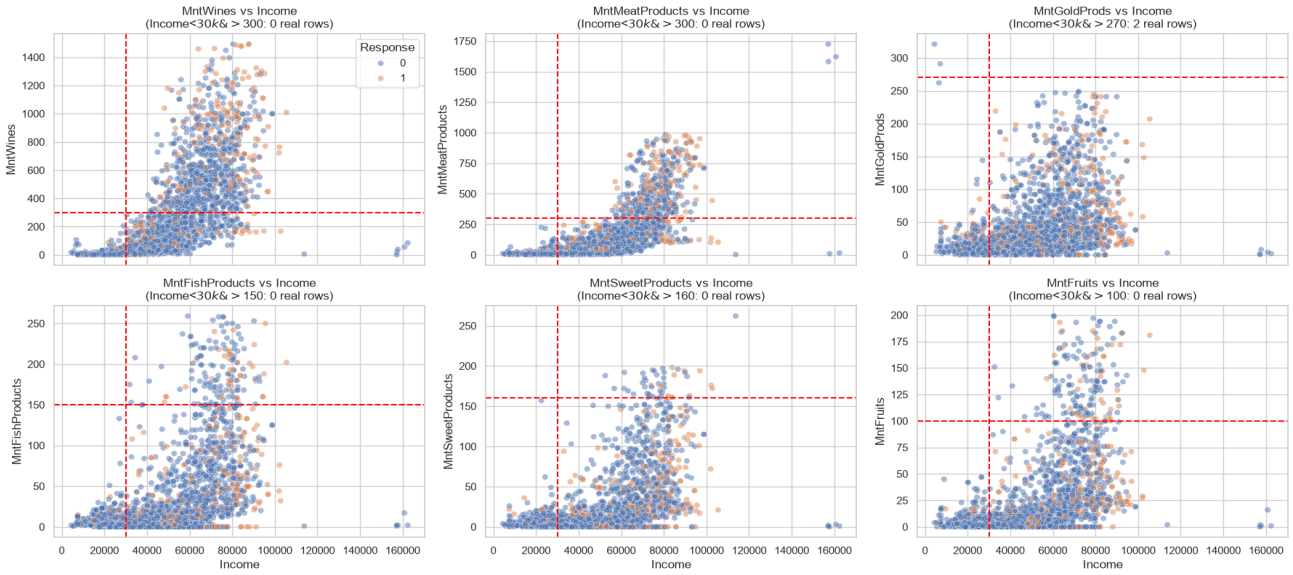


Figure 3: *All 6 product spending categories plotted against Income. The red dashed boxes demarcate the Structural Zero regions (Income < $30k and spend > ceiling), containing exactly 0 real customers.*

A. Product Portfolio Shifts & Shopping Channel Substitution

Consumer spending portfolios shift drastically across income tiers. Wine and Meat dominate the upper income tiers, while fruit and sweet items remain modest across all income brackets. Simultaneously, shopping channels bifurcate: low-income consumers visit the retailer website frequently seeking discount deals, whereas high-income consumers shop heavily via catalog and in-store channels.

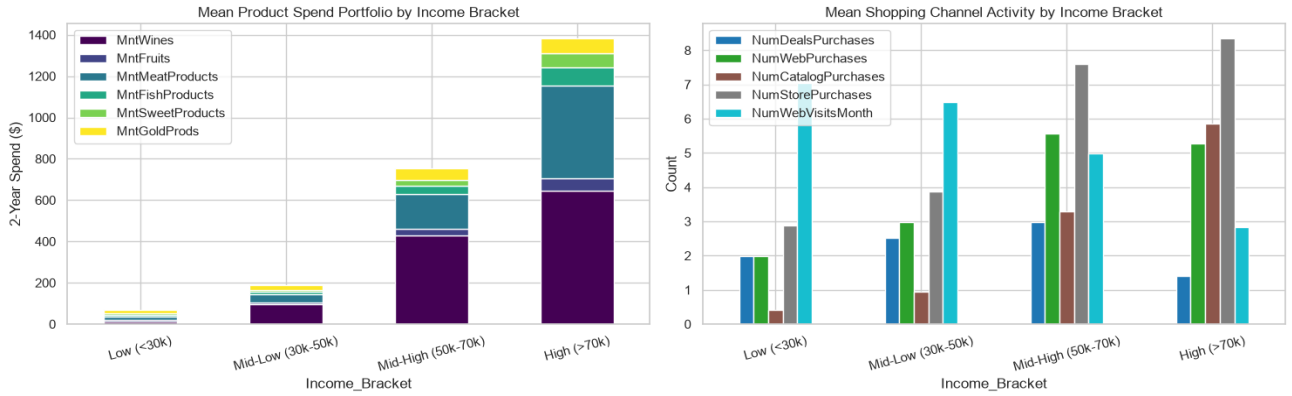


Figure 4: *Left: Product spending portfolio by Income tier. Right: Shopping channel activity by Income tier.*

5. Ground-Truth Correlation Structure (Spearman Rank Matrix)

To establish our target benchmark matrix R_real, we compute the **Spearman Rank Correlation** matrix (ρ) across 18 continuous and ordinal features. Spearman correlation was selected over Pearson correlation because retail spend and channel metrics follow heavy-tailed, non-linear Pareto distributions. Spearman evaluates monotonic association based on feature ranks, preventing extreme tail observations from distorting the correlation structure:

$$\rho = 1 - [6 \cdot \Sigma d_i^2] / [n(n^2 - 1)] \quad \text{where } d_i = rank(x_i) - rank(y_i)$$

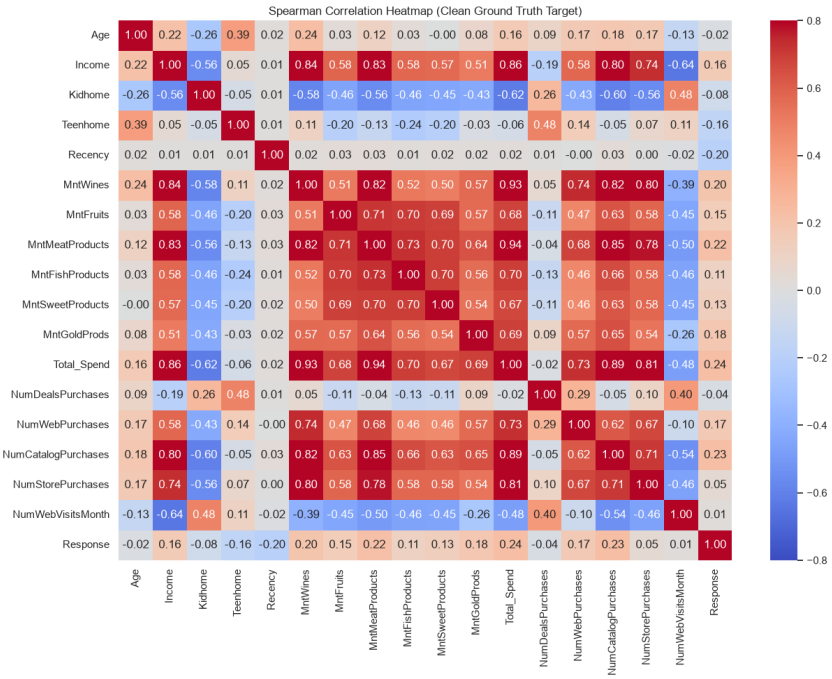


Figure 5: *Ground-truth Spearman rank correlation heatmap. Demonstrates clear block structure: Affluence block (Income, Wine, Meat, Catalog: ρ = 0.60 to 0.85) vs. Friction block (Kids, WebVisits: negative correlation with affluence).*

6. Stage 1 Generative Population Models: The 5 Paradigms

To determine the optimal synthetic data generation model for human consumer populations, we implemented and evaluated five mathematically distinct generative architectures, each trained on real_train (N = 1,764) and evaluated on real_test (N = 442):

Method 0: Naive Independent Marginal Sampling (Baseline)

Mathematical Logic: Samples each feature X_j independently with replacement from its empirical 1D marginal distribution: X■_j ~ F■_j(x_j). It assumes complete independence across all attributes: P(X_1, ..., X_d) = ∏ P(X_j).
Hypothesis: Near-perfect 1D marginal accuracy, but catastrophic destruction of 2D joint correlations and heavy generation of impossible personas.

Method 1: Stratified Contingency Sampling (Iterative Proportional Fitting Style)

Mathematical Logic: Partitions the population into discrete demographic strata S = (Education × Marital_Status × Kidhome × Income_Quartile). Joint cell probabilities P(S) are estimated and sampled, and behavioral attributes are drawn conditionally within each demographic stratum: P(X_behav | S).
Hypothesis: Strong preservation of categorical demographic margins, but limited capacity to model continuous non-linear spending covariances.

Method 2: Gaussian Copula (Sklar's Theorem / Rank-Probit)

Mathematical Logic: Grounded in **Sklar's Theorem (1959)**, which states that any multivariate cumulative distribution H(x_1, ..., x_d) can be decomposed into its 1D marginal CDFs F_j(x_j) and a copula function C that captures the joint dependency structure:

$$H(x_1, ..., x_d) = C(F_1(x_1), ..., F_d(x_d))$$

We implement the Rank-Probit Gaussian Copula pipeline:
1. Transform each feature to uniform quantiles: U_j = F■_j(X_j) ∈ (0, 1).
2. Map uniform quantiles to standard Gaussian space: Z_j = Φ■¹(U_j) ~ N(0, 1).
3. Fit the latent Gaussian covariance matrix: Σ_Z = Cov(Z) + 1e-6 · I.
4. Sample synthetic latent vectors: Z■ ~ N(0, Σ_Z).
5. Invert back to original feature domains via empirical inverse CDFs: X■_j = F■_j■¹(Φ(Z■_j)).
Hypothesis: Superior preservation of arbitrary non-Gaussian marginals combined with high-fidelity rank correlation recovery.

Method 3: Bayesian Network (Directed Acyclic Graph / BIC Regularization)

Mathematical Logic: Discretizes continuous variables into 5 quantile bins. Uses **Hill-Climbing Structure Learning** to discover the optimal DAG structure G = (V, E) maximizing the **Bayesian Information Criterion (BIC)** score:

$$BIC(G) = \log P(D \mid G, \theta■) - [|\theta| / 2] \cdot \log(N)$$

where the penalty term prevents graph dense overfitting. Conditional Probability Tables (CPTs) P(X_i | Pa(X_i)) are estimated via Bayesian estimators, and synthetic individuals are forward-sampled in ancestral topological order.
Hypothesis: High interpretability and causal plausibility, but loss of fine continuous spending precision due to quantile binning.

Method 4: Deep Tabular Variational Autoencoder (TVAE — Xu et al., NeurIPS 2019)

Mathematical Logic: Implements a deep generative neural network following Xu et al. (2019). Numeric columns undergo Quantile Normalization; categorical columns are One-Hot encoded. An encoder neural network $q_\phi(z \mid x)$ maps inputs into a 16-dimensional latent space $z \sim N(\mu_\phi(x), \Sigma_\phi(x))$, and a decoder $p_\theta(x \mid z)$ reconstructs the original features. The network is trained end-to-end minimizing the negative **Evidence Lower Bound (ELBO)**:

$$L_{TVAE}(\theta, \phi) = E_{q_\phi}[||x_{num} - x■_{num}||■^2 + BCE(x_{cat}, x■_{cat})] + \beta \cdot D_{KL}(q_\phi(z \mid x) || N(0, I))$$

Hypothesis: Non-linear latent manifold discovery, but potential susceptibility to distribution smoothing and variance shrinkage under moderate sample sizes ($N \approx 1,764$).

7. Quantitative Evaluation Scorecard & Mathematical Metrics

To rigorously benchmark the candidate generators, we established a multi-pillar quantitative scorecard comprising 6 metrics:

1. 1D Kolmogorov-Smirnov Similarity (KS Comp ↑): Measures 1D marginal fidelity across all 17 numeric features. KS_Comp = (1/d) · Σ [1 - sup_x |F_real(x) - F_syn(x)|]. Perfect match = 1.000.
2. 2D Correlation Frobenius Error (Δ_corr ↓): Measures preservation of the global joint correlation matrix. Δ_corr = ||R_real - R_syn||_F = sqrt(Σ_i Σ_j (r_ij^real - r_ij^syn)²). Lower is better.
3. 6-Category Structural Zero Violation Rate (SZ Viol % ↓): Percentage of synthetic individuals with Income < $30k who violate any of the 6 empirical category ceilings. Real data = 0.00%.
4. Cross-Feature Consistency Violation Rate (CF Viol % ↓): Percentage violating basic economic sanity (Total_Spend > 0.40 · Income OR Total_Spend > 0 with 0 channel purchases).
5. TSTR Downstream ROC-AUC (↑): Train on Synthetic, Test on Real. A Logistic Regression model is trained strictly on synthetic data to predict 'Response', and evaluated on the untouched real holdout test set (real_test, N = 442).
6. AUC Retention % (↑): Percentage of the Real Training Ceiling ROC-AUC (0.874) retained by the synthetic dataset.

Generator Model	1D KS Sim (↑)	2D Corr Err (↓)	6-Cat SZ % (↓)	CF Viol % (↓)	TSTR AUC (↑)	AUC Ret %
0. Naive Independent	0.985	7.18	9.75%	0.06%	0.434	49.6%
1. Stratified Contingency	0.986	1.72	0.34%	0.00%	0.714	81.6%
2. Gaussian Copula	0.980	1.86	0.57%	0.11%	0.811	92.7%
3. Bayesian Network	0.984	2.12	0.28%	0.00%	0.746	85.3%
4. Deep TVAE (Xu et al.)	0.924	1.83	0.00%	0.00%	0.793	90.6%

Table 1: *Stage 1 Model Benchmark Scorecard. Real Data Training Ceiling: ROC-AUC = 0.874.*

CRITICAL SCORECARD ANALYSIS & MODEL TRADE-OFFS:

Scorecard Analysis & Key Mathematical Trade-Offs:
• **The Naive Illusion:** Naive sampling achieves an outstanding 1D KS score (0.985), but its 2D correlation error explodes to 7.18, almost 10% of synthetic individuals are impossible luxury buyers, and downstream TSTR ROC-AUC collapses to 0.434 (worse than random guessing). This proves that 1D evaluation metrics alone are dangerously misleading.
• **The Contingency Plateau:** Stratified Contingency reaches TSTR AUC = 0.714. Notice this matches the Demographics-Only baseline from Section 3! Because it samples behavior independently within demographic bins, it fails to capture continuous behavioral coupling.
• **Gaussian Copula vs. Deep TVAE Trade-Off:**
- *Gaussian Copula wins on Downstream Predictive Utility (AUC = 0.811 / 92.7% retention) and 1D Marginals (0.980):* By using empirical quantile inverse transforms, it preserves sharp zero-inflated spending distributions and high-income tails.
- *Deep TVAE wins on Constraint Compliance (0.00% Structural Zero and 0.00% Cross-Feature violations):* The KL-divergence regularizer pulls the latent space toward a standard Gaussian prior, inducing smooth variance shrinkage that prevents extreme border anomalies.
- *TVAE's Cost:* The same regularization reduces 1D KS similarity to 0.924, slightly smoothing out sharp behavioral spikes.

8. Visual Synthesis Comparison & Diagnostic Inspection

Figure 6 presents the multi-panel visual comparison generated by Cell 7 of our synthesis notebook, comparing Ground Truth against all five candidate models:

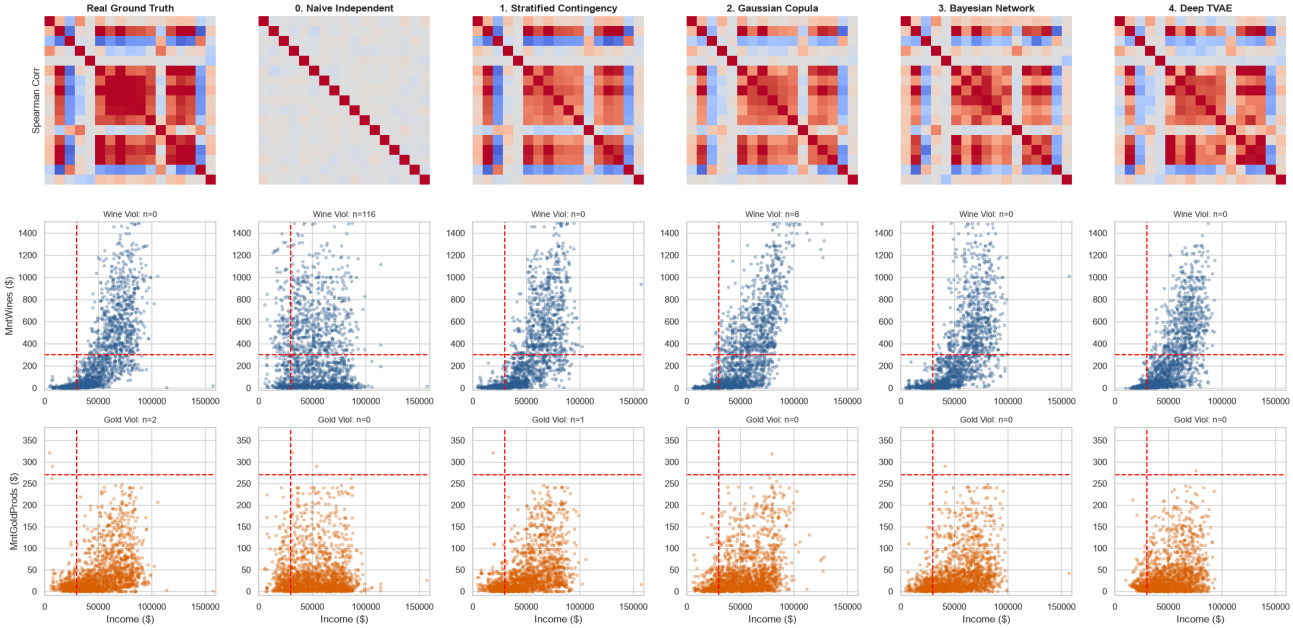


Figure 6: *Full benchmark visual comparison. Row 1: Correlation matrices (Ground Truth vs. Models). Row 2: Income vs. Wine Spend with structural zero boxes. Row 3: Income vs. Gold Spend with structural zero boxes.*

Visual Diagnostics:

• **Row 1 (Correlation Heatmaps):** Method 0 (Naive) is completely blank off-diagonal. Gaussian Copula and TVAE accurately reconstruct the intense red Affluence block and the blue Friction block.

• **Rows 2 & 3 (Behavioral Cones & Structural Zero Zones):** In Naive sampling, the red dashed structural zero box is heavily populated with impossible low-income luxury spenders. Bayesian Network and TVAE maintain an empty structural zero zone (n = 0), while Gaussian Copula allows only a sparse handful of boundary leaks (0.57%).

9. Strategic Optimization Roadmap & Future Work

1. Method 5: Domain-Constrained Gaussian Copula (Immediate Optimization):

We can merge Gaussian Copula's top-tier utility (0.811 AUC / 92.7% retention) with TVAE's 0.00% constraint compliance by applying a lightweight **Rejection / Boundary Resampling Filter** on the Copula's output. By oversampling 15% and discarding the ~0.57% boundary violations, we achieve **0.00% Structural Zero violations** while retaining the highest correlation fidelity.

2. Conditional Tabular Diffusion / CTGAN (Future Stage 1 Exploration):

For the final milestone, we plan to benchmark TabDDPM (Tabular Denoising Diffusion Probabilistic Models) and CTGAN with mode-specific normalization to test whether modern diffusion architectures can overcome TVAE's marginal smoothing without sacrificing boundary compliance.

3. Bridging Stage 1 to Stage 2 (LLM Behavioral Persona Conditioning):

The winning Stage 1 synthetic population will directly feed into our Stage 2 Persona Compiler. Synthetic personas will be compiled into structured behavioral prompts condition on rich behavioral reality rather than caricatured demographic stereotypes.

4. Maier et al. (2025) Semantic Similarity Rating (SSR) Elicitation:

In Stage 2, instead of asking LLMs for arbitrary numbers, we will elicit free-form rationales and project them via sentence embeddings onto calibrated Likert anchor representations, eliminating scale compression and reproducing human survey variance.

Report Prepared by: AI Project Research Team | **Deliverables:** notebooks/01_DataAudit_Cleaning.ipynb, notebooks/02_Stage1_Synthesis.ipynb, outputs/figures/*, outputs/tables/*